
Sums

Examples

Computing Average Values

and other formulas from statistics involve sums. Here are some examples, from statistics and communication theory.

Total Squared Error

If the data in the list x is supposed to be the same as the data in list y , then the total squared error is

$$\text{Out[]= total squared error} = \sum_{i=0}^n (x_i - y_i)^2$$

Shannon Entropy

From communication theory, if p is a probability distribution (in particular, $0 \leq p \leq 1$)

$$\text{Out[]= Shannon entropy} = \sum_{i=0}^n p_i \log(p_i)$$

Definition and Notation

Sigma notation Σ denotes a sum.

Usually there is a variable whose scope is only within the Σ expression, e.g.,

$$\text{Out[]= } \sum_{i=0}^5 i = 0 + 1 + 2 + 3 + 4 + 5$$

The variable can be any symbol, but is usually either i , j , or k . The choice of variable does not affect evaluation.

$$\text{Out[]= } \sum_{k=0}^5 k = 0 + 1 + 2 + 3 + 4 + 5$$

In sums the operator is Plus. There is a corresponding notation for Times and products using Π instead of Σ .

Sometimes the sum is taken to be infinite in which case the lower limit or the upper limit can be ∞ , e.g.,

$$\text{Out}[*]= \sum_{i=1}^{\infty} 1/i^2 = \pi^2/6$$

Infinite sums sometimes are not well-defined, though sometimes we can say they go to infinity.

$$\text{Out}[*]= \sum_{i=1}^{\infty} 1/i = \infty$$

Other notation is possible. For instance, it is sometimes convenient to use a set S instead of an index.

For instance if $S = \{2, 4, 8\}$ then

$$\text{Out}[*]= \sum_{i \in S} 1/i = 1/2 + 1/4 + 1/8$$

This can be convenient for sets which are not arithmetic sequences, e.g., the primes

$$\text{Out}[*]= \sum_{i \in \text{primes}} 1/i^2 \approx 0.452247$$

Recursion

Sums correspond to a recursion, e.g., set for $n \geq 0$,

$$\text{Out}[*]= f(n) = \sum_{i=0}^n a_i$$

then

$$f(n) = f(n-1) + a_n$$

Some recursions can be converted to sums by finding a summation factor.

For instance, in the tower of Hanoi problem $T(n) = 2T(n-1) + 1$, which is almost in the right form as the above identity. If we let $f(n) = 2^n T(n)$ then it is in the right form, and we immediately get that $f(n) = \text{sum } 2^n$ which is a geometric sum so it has a closed form.

Many of the tricks for sums also work for recursions.

Rules and Laws

The usual laws from algebra and addition apply.

Associative:

$$\text{Out}[*]= \sum_{i \in S} a_i + \sum_{i \in S} b_i$$

$$Out[*]= \sum_{i \in S} (a_i + b_i)$$

Distributive law:

$$Out[*]= \sum_{i \in S} \alpha a_i$$

$$Out[*]= \alpha \sum_{i \in S} a_i$$

The commutative law amounts to reordering the terms with a permutation σ :

$$Out[*]= \sum_{i \in S} a_i$$

$$Out[*]= \sum_{i \in S} a_{\sigma(i)}$$

A simple example is $S = \{1, 3, 5\}$

$$1 + 3 + 5 = 3 + 5 + 1$$

where σ is the permutation given by this table:

i	$\sigma(i)$
1	3
3	5
5	1

Tricks for summing

Simple Sums

If the terms are constant, then it is just multiplication.

$$Out[*]= \sum_{i=1}^n \alpha = n \alpha$$

e.g.,

$$Out[*]= \sum_{i=1}^{20} 1 = 20$$

Famous Sums

Harmonic Sums define the function $H(n)$. Famous, but no closed form. This function grows like $\ln x$.

$$\text{Out}[*]= \sum_{i=1}^n 1/i = H(n)$$

for instance

n	1	2	3	4	5	6
H(n)	1	$\frac{3}{2}$	$\frac{11}{6}$	$\frac{25}{12}$	$\frac{137}{60}$	$\frac{49}{20}$
decimal	1.	1.5	1.83333	2.08333	2.28333	2.45

The Harmonic numbers come up in various sums, but other famous sums have closed forms.

Geometric Sums: for $\alpha \neq 1$,

$$\text{Out}[*]= \sum_{i=0}^n \alpha^i = (1 - \alpha^{n+1}) / (1 - \alpha)$$

Arithmetic Sums boil down to

$$\text{Out}[*]= \sum_{i=0}^n i = n(n+1)/2$$

Telescoping Sums:

$$\text{Out}[*]= \sum_{i=0}^n a_{i+1} - a_i = a_{n+1} - a_0$$

Algebra

Algebra is often useful.

For instance, algebra can convert an expression into a simple telescoping sum:

$$1/(k(k+1)) = 1/k - 1/(k+1)$$

$$\text{Out}[*]= \sum_{k=1}^n 1/(k(k+1)) = 1 - 1/(n+1)$$

Or algebra might involve reordering terms. For instance if σ is the permutation that reverses the order of the range from 0 to n then $\sigma(i) = n - i$

$$\text{Out}[*]= \text{sum} = \sum_{i=0}^n i = \sum_{i=0}^n (n - i)$$

Hence sum + sum is

$$\text{Out}[*]= 2 \text{ sum} = \sum_{i=0}^n i + (n - i)$$

$$\text{Out}[*]= 2 \text{ sum} = \sum_{i=0}^n n$$

which is a constant sum so is

$$2 \text{ sum} = n (n + 1)$$

proving that this sum is $n (n + 1)/2$

Perturbation

This is a special case of using algebra.

Let's see an example and find the formula for the geometric sum. Assume $\alpha \neq 1$, this is what we want to derive:

$$\text{Out}[*]= g(n) = \sum_{i=0}^n \alpha^i = (1 - \alpha^{n+1}) / (1 - \alpha)$$

We split off the first term and on the other side of the equation split off the last term.

$$\text{Out}[*]= g(n) = \alpha^0 + \sum_{i=1}^n \alpha^i = 1 + \sum_{i=1}^n \alpha^i$$

$$\text{Out}[*]= g(n) = \sum_{i=0}^{n-1} \alpha^i + \alpha^n$$

Now since we can factor out an α , the two sums are comparable. Call that value $g(n - 1)$.

$$\text{Out}[*]= g(n) = 1 + \sum_{i=1}^n \alpha^i = 1 + \alpha \sum_{i=0}^{n-1} \alpha^i = 1 + \alpha g(n - 1)$$

and

$$\text{Out}[*]= g(n) = g(n - 1) + \alpha^n$$

Together we get that

$$\text{Out}[*]= 1 + \alpha g(n - 1) = g(n - 1) + \alpha^n$$

Finally solve for $g(n-1)$

Guess and Prove it

If we guess that

$$\text{Out[4]= } f(n) = \sum_{i=0}^n i = \frac{1}{2}n^2 + \frac{1}{2}n$$

then we check that $f(0) = 0$ is correct and $f(1) = 1/2 + 1/2 = 1$ is correct. Instead of continuing forever, we can rely on induction. To show the induction step, that $f(n+1) - f(n) = n+1$, we use algebraic manipulations. So it satisfies the correct recurrence is another way of proving it.

In[5]:= `f[n+1] - f[n] /. f[n_] => n (n+1) / 2`

$$\text{Out[5]= } -\frac{1}{2}n(1+n) + \frac{1}{2}(1+n)(2+n)$$

In[6]:= `Simplify[%]`

$$\text{Out[6]= } 1 + n$$

Guess the form and solve for parameters

Let's do this one again.

$$\text{Out[4]= } \text{sum} = \sum_{i=0}^n i$$

This time we guess that the answer is a quadratic polynomial, and then we solve for the coefficients.

$$\text{Out[4]= } f(n) = \sum_{i=0}^n i = \alpha_2 n^2 + \alpha_1 n + \alpha_0$$

With three variables we should need three equations. We can evaluate $f(n)$ to get as many equations as we need.

Out[4]=

n	f(n)	$\alpha_0 + n \alpha_1 + n^2 \alpha_2$
0	0	α_0
1	1	$\alpha_0 + \alpha_1 + \alpha_2$
2	3	$\alpha_0 + 2 \alpha_1 + 4 \alpha_2$
3	6	$\alpha_0 + 3 \alpha_1 + 9 \alpha_2$
4	10	$\alpha_0 + 4 \alpha_1 + 16 \alpha_2$
5	15	$\alpha_0 + 5 \alpha_1 + 25 \alpha_2$
6	21	$\alpha_0 + 6 \alpha_1 + 36 \alpha_2$

Take the first three

$$0 = \alpha_0$$

$$1 = \alpha_0 + \alpha_1 + \alpha_2$$

$$3 = \alpha_0 + 2 \alpha_1 + 4 \alpha_2$$

and we solve to get

$$\alpha_0 = 0$$

$$\alpha_1 = 1/2$$

$$\alpha_2 = 1/2$$

At this point we only know the formula works for the first three.

By checking the other known values, we increase our confidence in the formula.

But to actually prove it is right, we would need a proof. In this case the simplest proof involves plugging the formula into the recurrence and simplifying, which proves the induction step. If we also check the base cases then induction implies the formula is correct.

The only difference from the guess + prove technique is that we found the formula with linear algebra instead of guessing it.

Expand and Contract

The expand part of this method means we replace a single sum with a double sum. As a double sum, we perform legal operations, often switching the order of summation, to get something simpler. The contracting part is simplifying the double sum to the closed form.

Discrete Calculus

Generating Functions

Double Sums

Definition

In general, one could have a set of multi-indices to sum over, e.g., if $S = ((1,2), (3,5), (8,21))$ then

$$\text{Out[8]=} \sum_{i,j \in S} i/j = 1/2 + 3/5 + 8/21$$

or one could write a description of the indices under $\sum$, e.g.,

$$\text{In[11]:=} \text{Row[{\text{Column[{"", Style["\sum", 25], "1 \le i \le j \le 3"}], Alignment \to Center},$$

$$\text{" i/j", " = 1/1 + 1/2 + 1/3 + 2/2 + 2/3 + 3/3 = 9/2"}]}$$

$$\text{Out[11]=} \sum_{1 \leq i \leq j \leq 3} i/j = 1/1 + 1/2 + 1/3 + 2/2 + 2/3 + 3/3 = 9/2$$

More common is to use a double sum.

$$\text{Out[12]} = \sum_{i=1}^n \sum_{j=i}^n i/j = 1/1 + 1/2 + 1/3 + 2/2 + 2/3 + 3/3 = 9/2$$

The scoping of the iteration variable means that the limits of the inner sum can use the variable from the outer sum. In the above example, the inner sum starts at i .

In a finite sum the order doesn't matter, but in principle the outer variable is set first and then the inner sum is evaluated. Here it is color coded, evaluated from left to right:

$$\text{Out[17]} = \sum_{i=1}^n \sum_{j=i}^n i/j = 1/1 + 1/2 + 1/3 + 2/2 + 2/3 + 3/3$$

Out[18]=

i=1	1
i=2	2
i=3	3
j=1	1
j=2	2
j=3	3

In principle one can have triple sums, or as many as needed.

Changing the Order of Summation

If an infinite sum converges absolutely then its terms can be rearranged.

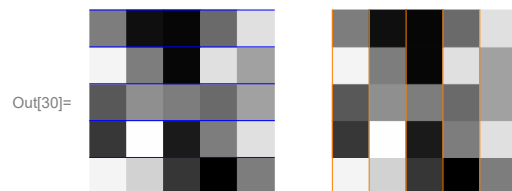
For a finite sum there is no concern about rearranging changing the result. It's always the same.

Converting it accurately can be tricky if the limits are not simple. This has a direct analogy with changing the limits for double integrals.

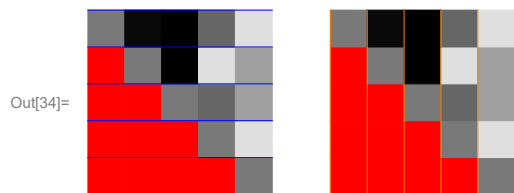
The simple case is

$$\text{Out[22]} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} = \sum_{j=1}^n \sum_{i=1}^n a_{ij}$$

Every index in the square is being sampled, and since the square is symmetric along $y=x$, the sum can go in either order. Every possible i,j index is covered exactly once either way. We can sum each row first, in blue, or sum each column first, in orange.



In the case where $j \geq i$, then it is like a triangle and the limits are different. In this picture the top left cell corresponds to $i=1, j=1$, and i corresponds to the row.



When i is the outer loop (on the left) then in each row, j ranges from i to n .

When j is the outer loop (on the right) then in each column, i ranges from 1 to j .

Out[32]=

$$\sum_{i=1}^n \sum_{j=i}^n a_{ij} = \sum_{j=1}^n \sum_{i=1}^j a_{ij}$$

Let's consider the color-coded sum again.

Out[34]=

$$\sum_{i=1}^n \sum_{j=i}^n i/j = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{2}{2} + \frac{2}{3} + \frac{3}{3}$$

Switching the order of summation gives a slightly different order

Out[35]=

$$\sum_{j=1}^n \sum_{i=1}^j i/j = \frac{1}{1} + \frac{1}{2} + \frac{2}{2} + \frac{1}{3} + \frac{2}{3} + \frac{3}{3}$$

Factoring

Factoring is the opposite process of expanding things out.

For example, expanding out $(a_1 + a_2 + a_3)(b_1 + b_2 + b_3)$

In[36]:= **Expand** [$(a_1 + a_2 + a_3) (b_1 + b_2 + b_3)$]

Out[36]= $a_1 b_1 + a_2 b_1 + a_3 b_1 + a_1 b_2 + a_2 b_2 + a_3 b_2 + a_1 b_3 + a_2 b_3 + a_3 b_3$

In[37]:= **Factor** [%]

Out[37]= $(a_1 + a_2 + a_3) (b_1 + b_2 + b_3)$

Generalizing the distributive law, expanding involves every possible product.

This is easier to write as a double sum.

$(a_1 + a_2 + a_3)(b_1 + b_2 + b_3)$ expanded out is

Out[38]=

$$\sum_{i=1}^3 \sum_{j=1}^3 a_i b_j$$

In general, the limits don't need to be the same, e.g.,

$$\text{Out[40]=} \sum_{i=1}^n \sum_{j=1}^m a_i b_j = \left(\sum_{i=1}^n a_i \right) \times \left(\sum_{j=1}^m b_j \right)$$

That's the main formula for factoring.

Note that when $n=m$ the operation on the left involves n^2 multiplications and additions. The operation on the right involves $2n$ additions and one multiplication.

So this kind of factoring, when possible, can be very efficient.

Also, when it happens, it can provide closed form solutions such as

$$\text{Out[44]=} \sum_{i=1}^n \sum_{j=1}^m 2^i/j = \left(\sum_{i=1}^n 2^i \right) \times \left(\sum_{j=1}^m 1/j \right) = (2^{n+1} - 2) H(m)$$